PROJECT TECHNICAL DOCUMENTATION

BioSpecInfo

Web platform for computational chemistry, physics and mathematics

Scientific Accuracy and Data Provenance

DOCUMENT CONTROL

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Scientific Accuracy and Data Provenance — BioSpecInfo

Field	Value
Software	BioSpecInfo
Author	Samuele Pio Provenzano
Version described	`bsi-v248`
Purpose	Document how the scientific data shown by the application are generated, by what method they are verified, and what the declared limits are.

> **Why this document exists.** A chemistry teaching application can be impeccable from a software point of view and wrong from a scientific one, and the user has no way of noticing: a plausible molecular weight, a convincing-looking spectrum and a wrong structure all present themselves in the same way. This document describes the checks that make that error **visible and automatic**, and declares what stays beyond their reach.

1. Principle: every scientific datum must be open to contradiction

The criterion adopted throughout the application is that a scientific datum is not considered verified because it *looks* right, but because **a second, independent datum exists that would confirm or refute it**. Where that second datum does not exist, the limit is declared instead of being filled with an estimate presented as a measurement.

Three concrete applications of this principle:

Domain	First statement	Second, independent statement	Check
Drug database	structure (SMILES)	molecular weight from the literature	the weight recomputed from the structure must agree
IR / NMR spectra	functional groups recognised	molecular structure via RDKit	groups are sought on the graph, not on the text of the SMILES
Physical constants	tabulated value	symbol and unit of measure	the lookup refuses ambiguous matches instead of picking one

2. Pharmacological database — 263 entries

2.1 The verification method

Every entry declares **two** properties of the same molecule: the structure, in SMILES notation, and the molecular weight. They are independent statements — the first describes atomic connectivity, the second is a number taken from the literature — and therefore comparable:

```
weight computed by RDKit from the structure   ↔   weight declared in the table
```

A discrepancy above **0.6 u** (a tolerance that covers rounding and small differences between atomic-mass tables, but not a real error, which is worth tens of units) indicates that at least one of the two statements is wrong.

The check is automated (`tools/verifica-farmaci.js` , runnable by anyone holding the repository) and requires no human judgement: it does not ask anyone to trust a source, it compares two numbers.

2.2 What it found

Applied to the 143 pre-existing entries, the check identified **89 errors**. The distribution is significant: the **molecular weights were correct** — taken from the literature — while the **structures were wrong**.

Drug	Structure declared
Ketamine	an unrelated pyrrolidinone
Morphine	non-matching structure (in two separate entries)
Lorazepam	missing the 3-position hydroxyl that defines its identity
Atorvastatin	missing the anilide group
Testosterone	missing the Δ^4 double bond of ring A

Plus **7 duplicated drugs** in different categories, with data that sometimes diverged between the two copies.

2.3 The discrepancy as a diagnostic instrument

The numerical value of the difference is not merely an alarm: it indicates **which** fragment is missing or superfluous.

Δ (u)	Interpretation	Real case
28.05	C_2H_4	ketamine, fentanyl, morphine
16.00	one oxygen atom	tetracycline (one hydroxyl too many)
14.02	a CH_2 group	doxycycline
2.02	a missing double bond	testosterone (Δ^4)

Δ (u)	Interpretation	Real case
0.98	cytosine ⟷ uracil	**sofosbuvir**

The last case deserves a note: a discrepancy of 0.98 u corresponds exactly to replacing an −NH group with an oxygen atom, that is, to the difference between cytosine and uracil. Sofosbuvir is a **uridine** analogue, and the proposed structure contained the wrong base. The number showed where to look.

2.4 Current state and declared limits

Drugs in the database	**178** (it was 143)
With a verified structure	**157** (it was 153)
Defects	**0**
Declared and accepted deviations	**21** (it was 25)

Verification is **conformant**: no entry presents a structure that contradicts its own molecular weight. The 21 deviations are entries **without a structure**, each recorded with its own reason in `../evidence/deviazioni-note.json` and reported in the verification report.

Defect versus deviation. The check distinguishes two situations that must not be confused:

- a structure that **contradicts** its own weight is a defect, and fails

verification;

- an **absent** structure may be an acceptable deviation — but only if

explicitly recorded, with its reason. An entry without a structure and **not** recorded fails verification like any other defect.

Recording a deviation is therefore a deliberate decision, traced in git and visible in the report, not a way of silencing a check.

And it must also be **revoked** when it is no longer needed. From version `bsi-v188` the bench also fails in the opposite case: an entry listed in the registry that **does** have a verified structure is a permission left switched on for nothing, and tomorrow it would silently cover a wrong structure put in its place.

2.4-bis Four entries that left the registry

Of the six entries without a structure, four were closed by taking the structure from **ChEMBL** — a source independent of this project — and verifying it with the same structure ⟷ molecular weight comparison that applies to all the others.

Entry	ChEMBL	Formula	Weight declared	Weight computed
Digoxin	`CHEMBL1751`	$C_{41}H_{64}O_{14}$	780.94	780.95
Vincristine	`CHEMBL90555`	$C_{46}H_{56}N_4O_{10}$	824.96	824.97
Tacrolimus topical (Protopic)	`CHEMBL269732`	$C_{44}H_{69}NO_{12}$	804.02	804.03

Entry	ChEMBL	Formula	Weight declared	Weight computed
Tacrolimus systemic (Prograf)	`CHEMBL269732`	$C_{44}H_{69}NO_{12}$	804.02	804.03

The reason recorded for digoxin said "every structure tried departs by 14-30 u from the literature weight": the declared weight was right, it was the structures tried that were wrong. That the comparison is real and not complacent can be seen by adding a single carbon to the correct structure — the bench reports **Δ 14.04** and fails.

The two that remain are not single molecules, and no SMILES notation represents them:

- **Ivermectin** is a mixture of homologues: at least 80 % B1a ($C_{48}H_{74}O_{14}$) and no

more than 20 % B1b ($C_{47}H_{72}O_{14}$). The declared weight, 875.10, is that of the B1a component alone. Showing its structure would pass off the majority homologue as the whole drug.

- **Artemether/Lumefantrine (Coartem)** is a combination of two distinct active

ingredients.

In neither case is a structure missing: there is no single one to show.

Group	Entries	Nature
D-01 — not single molecules	2	ivermectin (mixture of B1a/B1b homologues) and artemether/lumefantrine (combination of two active ingredients). It was six: the other four were closed with the structure from ChEMBL, see §2.4-bis.
D-02 — not representable in SMILES	19	monoclonal antibodies, proteins and peptides (trastuzumab, pembrolizumab, insulin, semaglutide, ciclosporin A…). The absence is correct, not a gap.

> The choice is deliberate: a wrong datum replaced by another wrong datum is no progress, and a declared absence is preferable to an unverified presence.

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3. Spectral prediction

3.1 Structural, not textual, recognition

Functional groups are identified with **SMARTS patterns applied to the molecular graph** through RDKit (`bsi-spettri.js`), not with regular expressions applied to the SMILES string.

The difference is not stylistic. The earlier textual method had two structural defects:

1. **Exclusivity.** The checks were organised in `else if` chains, so **only one

functional group per molecule** was recognised.

1. **Blindness to structure.** A regex reads characters, not bonds.

The emblematic case is **acetylsalicylic acid** (`CC(=O)Oc1ccccc1C(=O)O`), which contains both an ester and a carboxylic acid. Textual recognition identified only the ester; the resulting IR spectrum lacked the broad O–H band at ~3000 cm^{-1} and the acid C=O at ~1690 cm^{-1} — the two bands that identify the molecule — and the ^1H NMR spectrum lacked the carboxylic proton signal at 11.6 ppm.

Graph-based recognition is **additive by construction**: a molecule may present every group it possesses.

3.2 Representation conventions

Axis conventions are not uniform across techniques, and are verified automatically (`test_assi.js`, `test_assi_canvas.js`) by extracting the tick labels from the drawing produced:

Technique	Abscissa	Direction	Ordinate
IR	wavenumber (cm^{-1})	decreasing, 4000 → 400	**% transmittance**, bands pointing down
^1H NMR	δ (ppm)	decreasing, 14 → 0	intensity, TMS reference at 0
UV-Vis	λ (nm)	increasing, 200 → 400	absorbance
MS	m/z	increasing	relative abundance

3.3 Determinism

No spectrum contains random components. Earlier versions added pseudo-random noise to the trace on every redraw, with the consequence that the same molecule produced a different spectrum each time and two spectra were not comparable. A computed profile must be **reproducible**: where a realistic baseline is needed, a deterministic function of the wavenumber is used.

Likewise, in the NMR predictor the peak heights were generated at random. In a ^1H spectrum the height **is** the quantitative information — the integration — and inventing it is worse than omitting it. Currently:

- where the number of protons is known, the height is the real integration and

the label reports it (e.g. `7.26 (4H)`);

- where the prediction is a **range** (e.g. "6.5–8.5 ppm"), the band of the range

is drawn with a mark on the typical value and a uniform height: the chart shows *where* the signal falls without asserting how intense it is.

3.4 Chemical effects modelled

Effect	Implementation
Carbonyl conjugation	a ~25 cm^{-1} lowering applied **to the individual carbonyl**, not globally
Phenolic ester	raising to ~1760 cm^{-1}
C–H intensity	proportional to the number of CH_2/CH_3 groups present

Effect	Implementation
Aromatic ring substitution	"out of plane" bands 900–690 cm^{-1} distinguished for ortho/meta/para/mono
Aldehyde Fermi doublet	bands at 2820 and 2720 cm^{-1}
Amide bands I and II	1655 and 1550 cm^{-1}

Verification: acetylsalicylic acid produces an aryl ester at 1760 cm^{-1} and a conjugated acid at 1690 cm^{-1} — the two values that identify it — while acetophenone (conjugated ketone, 1690) and ethyl acetate (unconjugated ester,

1. confirm that the effect is applied selectively and not indiscriminately.

3.5 Declared limits

> The spectra produced are **predictions based on functional-group tables**, not measured spectra nor quantum-mechanical simulations. They indicate where the bands of the groups present fall, with tabulated positions and intensities. **They do not replace an experimental database** (SDBS, NIST) nor an instrumental measurement, and the chart says so.

In particular the following are not modelled: second-order spin-spin couplings, solvent and concentration effects, combination bands and overtones, solid-state polymorphism.

3-bis. Cheminformatics and QSAR models

3-bis.1 Where the methods come from

None of the methods in the **Cheminformatics** section was invented for the occasion. They are the ones in use, with the references that define them.

Method	Reference	How it is implemented here
Tanimoto coefficient	Rogers & Tanimoto, 1960	Bits in common over bits in union, with the 0/0 = 1 convention declared in the code
Morgan / ECFP fingerprints	Rogers & Hahn, *J. Chem. Inf. Model.* 50 (2010) 742	From RDKit MinimalLib, radius 2, 2048 bits
MACCS keys	Durant *et al.*, *J. Chem. Inf. Comput. Sci.* 42 (2002) 1273	From RDKit MinimalLib, 166 keys
Rule of five	Lipinski *et al.*, *Adv. Drug Deliv. Rev.* 23 (1997) 3	Over the RDKit descriptors, with the violation count
Veber filter	Veber *et al.*, *J. Med. Chem.* 45 (2002) 2615	Rotatable bonds ≤ 10, TPSA ≤ 140 $Å^2$
QED	Bickerton *et al.*, *Nat. Chem.* 4 (2012) 90	**Recomputed**: MinimalLib exposes 43 descriptors but not QED. The desirability-function parameters are those of the paper

Method	Reference	How it is implemented here
Butina clustering	Butina, *J. Chem. Inf. Comput. Sci.* 39 (1999) 747	Threshold on the Tanimoto distance, leaders chosen by neighbour count
Bemis–Murcko scaffolds	Bemis & Murcko, *J. Med. Chem.* 39 (1996) 2887	Ring systems plus the bonds that connect them
PAINS	Baell & Holloway, *J. Med. Chem.* 53 (2010) 2719	A subset of the SMARTS patterns, with the reason for the flag
Brenk alerts	Brenk *et al.*, *ChemMedChem* 3 (2008) 435	Likewise
SALI	Guha & Van Drie, *J. Chem. Inf. Model.* 48 (2008) 646	Δactivity / (1 − Tanimoto), over the pairs above the similarity threshold

3-bis.2 The four things that make a QSAR honest

A QSAR model is extremely easy to make look good. The four precautions below are what separate a number from a measurement, and all four are verified by the `test_cheminfo` bench.

Scaffold split. With a random split, close analogues from the same series end up on both sides: the model recovers what it has already seen. The scaffold split groups the molecules by Bemis–Murcko scaffold and assigns **whole scaffolds** to one side only. The R^2 that comes out is lower — and it is the one that survives the new molecule.

Null model by scrambling. The same model is retrained eight times on shuffled labels. If the true score does not exceed the best of the twins, the model has learnt nothing transferable, and the panel says so in those words. It is the check most often missing: without it, an R^2 of 0.4 over thirty molecules is indistinguishable from chance.

Applicability domain. A prediction on a molecule far from everything the model has seen is an extrapolation, not a prediction. The distance to the nearest training neighbour is shown next to every predicted value.

Scaffold-grouped cross-validation. A single split, over a small set, is a noisy number: on 28 molecules at 25 % the test set holds seven, and moving one shifts the R^2 by tenths. The folds group by scaffold — no scaffold sits in two folds — and the result is given as **mean ± deviation**, not as a single figure.

> **How much this matters, measured on the 28-inhibitor example.** The single split gives R^2 **0.861**. Five-fold validation gives **0.605 ± 0.452**, with the individual folds at 0.861 · 0.538 · 0.795 · **−0.150** · 0.980. The single split had drawn the lucky fold. Together the two figures say what neither says alone: the model works, but it depends appreciably on which molecules it gets for training — and on a set of twenty-eight molecules that is exactly what one should expect.

3-bis.3 Declared limits

Limit	Consequence
The example sets are **small** (28 and 40 molecules)	They exist to show the method, not to produce a usable model. On sets that size, the null model is the only serious defence — which is why it is there
The model is **linear in the kernel** (kernel ridge, Tanimoto kernel) or logistic	No neural network, no gradient boosting: on a few hundred molecules these give no demonstrable advantage, and the cost would be a model one cannot inspect
PAINS are not a verdict	A PAINS pattern signals that the compound has often been a false positive in fluorescence assays, not that it is inactive. The panel writes this next to every flag
The **PCA is on the descriptors**, not on the fingerprints	On binary fingerprints PCA is not very informative; loadings on descriptors can be read, and are shown
There is no **large-scale substructure search**	The workbench is sized for the sets one pastes in by hand, not for a library of millions of compounds

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3-bis.4 Two surfaces, two roles — and a number that was false

The application has **two** places where cheminformatics happens, and they serve different purposes. The distinction is written here because an assessor who opens the wrong page first would form the wrong impression.

	`rdkit_lab.html` — "RDKit Lab"	Cheminformatics section
Who it is for	Someone studying: one molecule at a time, to see what RDKit can say about it	Someone working: a set of molecules, to derive a defensible model
Input	One SMILES	A pasted set, with the measured activity
QSAR	**Empirical** estimates from the descriptors, declared as such in the panel	A model **trained** on the supplied data, with scaffold split, cross-validation and null model
Output	What is on screen	CSV and method report

> **A number that was false, and how the bench caught it.** The lab's "Pharma Pro" panel compared the molecule against eight reference drugs and displayed a similarity percentage. That percentage did not come from a fingerprint: it came from **eight bits of thresholded descriptors** — "has aromatic rings", "HBA > 4", "weight between 200 and 500" — over which a Tanimoto was computed. Measured: **caffeine against metformin gave 0.75**, while on Morgan fingerprints it is **0.024**. Two molecules with no fragments in common, shown at 75 %, on a page called "RDKit Lab" where RDKit was already loaded.
>
> Alongside that, three of the eight reference fingerprints were **written by hand** and had the "aromatic" bit wrong (caffeine, which has two aromatic rings, morphine and amoxicillin which have one), and the **SMILES for omeprazole was not omeprazole**: `COc1ccc2[nH]c(=S)cc2c1OC` is a structure RDKit rejects.

> All corrected: the comparison uses the Morgan fingerprint with the shared engine's Tanimoto, the fingerprints are computed rather than typed in, and omeprazole is omeprazole (monoisotopic mass 345.11, three aromatic rings — both verified). The two functions of the fake similarity were **removed**, not bypassed: leaving them around would have put them back in use at the first edit, and the bench checks that they no longer exist.

One implementation of Tanimoto. The lab had its own, which agreed with the engine's on ordinary cases but answered 0 where the engine answers 1 (two empty fingerprints). The page now loads `bsi-cheminfo.js` and calls that: `test_cheminfo` compares the two routes over all 28 reference pairs and demands a deviation of **exactly zero**.

3-ter. Alternative text for figures is not invented

The biochemistry guide carries **98 figures**. On replacing it, the accessibility bench rejected it: none had an `alt` attribute.

The easy way out was `alt=""`, and the bench would have accepted it — the comment in its code says, correctly, that for a **decorative** image this is the right thing: it tells the screen reader to skip it. But these are biochemistry figures, that is, information. Marking them decorative would have passed the check by telling screen-reader users to ignore 98 diagrams: a hollow green, of the same family as the "contrast 0" that did not cover the new panels (§3-bis) and the 100 % coverage that summed the wrong ranges.

The honest way was available: every image sits inside a `<figure>` with its own `<figcaption>`, and **the captions were written by the document's author**.

Figures	62, holding 98 images
Single-image figures	26 — the alt is the caption, minus the "Figura N." label
Two-panel figures	36
Pair captions split between the two panels	**33 out of 36**
Pairs sharing one caption	3 — they do not split unambiguously

All 36 pair captions distinguish the two panels with "a sinistra" (left) and "a destra" (right): where the split is unambiguous, each image receives its own half of the sentence. Where it is not, the two images share the whole caption — verbose to listen to, but **it cannot be wrong**, and a wrong description of a figure is worse than a verbose one.

> **Why not attempt a more aggressive split.** The remaining 3 could have been squeezed out with looser regular expressions. A split that goes wrong produces "this image shows X" next to an image that shows Y, and the screen reader reads it with the same confidence as the correct ones. The fallback is noisy; the error is invisible.

Measured after the change: **98 out of 98 with alternative text, none empty**, contrast **0** over 1,897 text elements.

3-quater. From prototype to working tool

The functions described in §3-bis were enough to say something defensible about a set of molecules. They were not enough to **work** with it: on day one, a cheminformatics group asks to search a substructure across the whole set, to see the SAR table, and to know what happens to activity when a chlorine is replaced by a methyl.

3-quater.1 What the library had, and what it does not

Before designing anything, MinimalLib 2025.03.4 was **probed at runtime** rather than assumed. Probing first changed the design: three of the capabilities below had gone unused because nobody knew they were there.

Present	What it is for
`SubstructLibrary`	Substructure search with a pattern fingerprint: it discards in bulk the molecules that cannot match, instead of attempting isomorphism on each
`get_rxn` · `Reaction.run_reactants`	Chemical transformations. This is what makes it possible to **cut** a bond, and from there come matched pairs and the SAR table
`generate_aligned_coords`	Aligning molecules to a common core
`get_molblock` · `get_v3Kmolblock`	**SDF** export, the format in which sets are exchanged between groups and programs

Absent	Declared consequence
`cleanup`, `neutralize`	Standardisation stops at the largest fragment and the canonical SMILES: **charges are not neutralised**
`canonical_tautomer`	Two tautomers written differently remain two distinct entries
`FragmentOnBonds`, `RWMol`	No direct molecule editing: cutting goes through a reaction
Maximum common substructure (MCS)	A SAR table's core must be **supplied**, not deduced. The panel offers eight ready cores and a free field

3-quater.2 The cut, and why two rules are needed

A reaction SMARTS breaks a bond and marks both ends with a dummy atom. Verified at runtime: `CCOc1ccccc1` gives three cuts under the fine rule (CH_3–CH_2, CH_2–O, O–aryl) and two under the coarse one; benzene and methane give none.

Rule	Cuts	When it is needed
fine (default)	Any acyclic single bond, **including terminal substituents** Cl, CH_3, OH, F	SAR table and fine transformations (Cl → CH_3)

Rule	Cuts	When it is needed
coarse (BRICS-style)	Excludes terminal atoms	Large sets, where fine cuts are too many; produces larger variable parts ("chlorophenyl → phenyl")

> **The fine rule was born of a defect.** The first draft used only the coarse one, and the SAR table of the test series **lost the chlorine on the ring**: it found the substituent on the nitrogen and missed exactly the column that matters, because chlorine is terminal. The fine rule is a superset of the coarse one: every cut the coarse rule finds, the fine one finds too.

3-quater.3 Matched molecular pairs

Two molecules form a pair when, cutting one bond in each, they are left with the **same context** and two different variable parts. The activity difference is attributed to that substitution and to nothing else — which is why this reading beats a correlation over the whole set, which averages over molecules differing in ten ways.

Verified on a series **built with known effects**: if the analysis does not recover them, it is not usable on real data.

Transformation	Expected	Found	Concordant pairs
chlorophenyl → phenyl	−1.0	**−1.0**	2 of 2
methylphenyl → phenyl	−0.5	**−0.5**	2 of 2
methyl → chlorine	+0.5	**+0.5**	2 of 2
ethylamide → propylamide	+0.2	**+0.2**	3 of 3

On the 28 example inhibitors, taken from ChEMBL: **449 pairs, 98 transformations** seen at least twice, 199 seen once, in 133 ms. At the top, **CF_3 → SO_2NH_2 with a median Δ of −2.65** over two concordant pairs.

Two declared choices:

- the **direction is normalised** on the alphabetical order of the two parts.

Without it the same effect would appear twice, with opposite medians, and neither would carry the right pair count;

- transformations seen **fewer than twice** are excluded from the table and

counted separately. A transformation seen once with Δ = +3 is not a discovery: it is an anecdote, and putting it at the top would have decisions made on it.

> **The "concordant" column is the one that decides whether to trust it.** A median of +1.0 over three pairs one of which is −2.0 is not the same as +1.0 over three pairs all positive, and the median alone does not say so.

3-quater.4 The SAR table, and how the position is found

The hard part is not finding the substituents: it is knowing which **position** each occupies. With no MCS and no molecule editing, the solution exploits a property of the cut: **the fragment containing the core carries the dummy atom exactly at the attachment point**. Matching the core onto that fragment reveals which of its atoms the dummy hangs off, and therefore which position the substituent occupies.

This also yields a necessary and free filter: if the dummy is **not** attached to a core atom, the cut happened inside a substituent and that piece is not the whole substituent. It is discarded.

On the 28 inhibitors, with a benzamide core: **20 molecules with the core, 2 positions, 8 excluded**. The excluded ones do not appear in the table: showing them with empty cells would make them look like part of the series with substituents that were not found.

> **A naming defect, and how it showed itself.** `scaffoldMurcko()` returned its result in a field called `smiles`, but that value **is not a SMILES**: it is a canonical fingerprint of the scaffold, built so that two molecules with the same scaffold produce the same string. It serves to **group** — folds, scaffold split — not to query. The SAR panel took it for a SMILES and proposed `6,6,6,6,7,…|0-13:1,…` as the core; decomposition answered "core not interpretable".
>
> The field is now called `chiave` (key) and declares `eUnoSmiles: false`; `smiles` remains as an alias. The bench pins that equal scaffolds give the same key, that different scaffolds give different keys, and that the key **does not read back as a SMILES**. A field's wrong name is a defect like any other.

3-quater.5 The rigour an assessor demands

Five things a modelling group puts in every report, and whose absence is the first objection in a meeting.

	Method	Why
Model comparison	Trivial reference (predict the mean), kNN over Tanimoto, kernel ridge — all on the **same folds**, grouped by scaffold	Comparing a model evaluated on one split with another evaluated on a different split is not a comparison. And the trivial reference sits at the top of the list, because it is the number every other must beat
Nested validation	Hyperparameters (k, lambda) are chosen on folds **internal to the training set**	Trying ten of them on the test set and reporting the best inflates the score: that number is not a prediction, it is the maximum of ten attempts
Conformal intervals	Split-conformal, quantile $\lceil (n+1)(1-\alpha) \rceil$ of the absolute residuals on a calibration portion split **by scaffold**	"7.2" and "7.2 ± 0.3" are two different pieces of information, and only the second says whether synthesis is worth it. Calibrating on analogues of the training set would give intervals too narrow precisely on new molecules

	Method	Why
Enrichment	EF at 1 %, 5 %, 10 % with the **maximum achievable** alongside; BEDROC (Truchon & Bayly, *J. Chem. Inf. Model.* 47 (2007) 488)	In a screen one does not look at the whole list. A ROC-AUC of 0.80 can hide a top-of-list with not a single active, because AUC also rewards the ordering in the tail, which nobody will buy
Learning curve	Score against training-set size, with the test set **whole** at every point	It answers the only question that matters when the model is mediocre: more chemistry or more data?

The comparison's verdict is proven in both directions, like the null model. On a real signal — activity proportional to halogen count, over ten distinct cores — kernel ridge reaches R^2 0.943 with a margin of 0.943 against a dispersion of 0.035, and the verdict is *beats it*. On the **same molecules with random labels** the best stops at −0.265, margin −0.034, and the verdict is *does not beat it*. A guard that never fires and one that always fires are the same defect.

Enrichment is verified on the three cases where the right value is computable by hand:

Ordering (2 actives of 10)	AUC	EF@20 %	BEDROC
Perfect: both actives on top	1	**5** — and it is the maximum achievable	**1**
Worst: both actives last	0	0	**0**, not a negative
Actives at ranks 2 and 5	**0.75** = 12/16	**2.5** = 1/0.4	0.119

The worst case is the one that counts: without it one would not know whether BEDROC reaches zero or goes below, and a negative BEDROC is the sign that the formula's additive term is missing.

3-quater.6 Limits of this block

Limit	Consequence
Pairs grow with the **square** of the molecule count	Measured: 6 molecules 45 ms, 40 molecules 227 ms with 1,255 pairs. Beyond 300 molecules the panel refuses to start and says so, instead of freezing the page
The cut is **single**	A pair differing by two substitutions at once is not found. This is the limit of the single-cut scheme, shared with the standard tools
The same transformation appears in **several forms**	Cuts at different positions give different contexts for the same chemical change (`*CC>>*CCC`, `*NCC>>*NCCC`, `*C(=O)NCC>>*C(=O)NCCC`). This is expected redundancy in the single-cut scheme, not an error
Charges are **not** neutralised and tautomers not canonicalised	MinimalLib does not allow it (§3-quater.1). A set mixing neutral and ionic forms of the same molecule counts them as distinct entries
The SAR table's core must be **supplied**	No MCS available. The panel proposes nothing automatically, because a wrong proposal is worse than no proposal

3.6 Per-atom NMR prediction: the source and the measured deviation

The $^{1}H/^{13}C$ predictor that assigns every signal to an ATOM (`bsi-nmr.js`) contains no numbers. The estimation tables live in **`bsi-pretsch.js`**, transcribed line by line from:

> E. Pretsch, P. Bühlmann, M. Badertscher, *Structure Determination of Organic Compounds — Tables of Spectral Data*, 4th ed., Springer. §4.1 pp. 82-84 · §4.5 pp. 100-102 · §5.1 p. 170 · §5.2 pp. 178-179 · §5.3 p. 182 · §5.5 pp. 188-189

They sit in a separate file so they can be **checked against the printed page** without reading the code that uses them, and so a wrong line is fixed in one place only.

scheme	formula	rows
monosubstituted benzenes, ^{13}C	$\delta = 128.5 + \Sigma Zi$	91
monosubstituted benzenes, ^{1}H	$\delta = 7.34 + \Sigma Zi$	66
substituted ethylenes, ^{1}H	$\delta = 5.25 + Zgem + Zcis + Ztrans$	42
substituted alkanes, ^{1}H	$\delta = base(CH_3/CH_2/CH) + \Sigma Z\alpha + \Sigma Z\beta$	31
terminal alkynes, ^{1}H	direct value per substituent	30
aliphatics, ^{13}C	$\delta = -2.3 + \Sigma Zi + \Sigma Sj$	24
steric corrections Sj	by degree of the observed C and of the α atom	4×4

The deviation is measured, on molecules that did not choose the parameters. The test molecules are split into two sets: *tuning*, used to choose values and rules, and *validation*, never used for that. The figure declared in the application's panel is the validation one.

	measured mean deviation
^{13}C, tuning (9 molecules)	0.71 ppm
^{13}C, validation (22 molecules)	**0.92 ppm**
^{13}C, worst case	4.9 ppm (cyclohexanone)
^{1}H (14 molecules)	0.06 ppm
^{1}H, aromatics only	0.03 ppm

The bench (`test_nmr`, 56 checks) also requires validation to stay **worse** than tuning: were they to become equal it would mean a molecule had been moved from one set to the other, and the figure would no longer say anything.

What is still wrong has not been corrected after the fact. Biphenyl misses its ipso carbon by 4.6 ppm, because the table's phenyl increment ($Z_1 = 8.1$) does not describe it. That number **has not been adjusted**: changing a transcribed value because a validation molecule does not fit is exactly how the 0.92 ppm declared above would become a lie.

Declared limits. This is not a quantum calculation: it does not predict solvent effects, does not distinguish conformers (the source's conformational corrections K are transcribed but evaluate to 0, because conformation cannot be derived from a SMILES) and does no two-dimensional NMR. Coupling constants come from a table of **typical** values by geometric rela-

tionship, not calculated. **Substituted heteroaromatics** have, in this source, no positional increment table as benzenes do: they remain the module's least reliable prediction, and the elucidation tool declares it by lowering the confidence when it meets one.

3.7 Two-dimensional NMR: COSY, HSQC, HMBC

`bsi-nmr2d.js` holds no shifts: it asks the per-atom predictor for them and builds the correlations by **walking the graph**. This is why it could only be done once the Pretsch tables arrived: it needs the shift of *every* proton and *every* carbon, plus the index of the atom carrying them.

map	topological rule
HSQC	proton and carbon **one** bond apart (^{1}J); inverted sign on CH_2
COSY	two protons **three** bonds apart (H–C–C–H), excluding equivalents
HMBC	proton and carbons **two or three** bonds away (^{2}J, ^{3}J)

Exchangeable hydrogens (O–H, N–H) give no off-diagonal spots: they exchange with the solvent too fast for the coupling to be seen. They do appear on the COSY diagonal, because they are protons like any other.

Verified by counting paths (`test_nmr2d`, 40 checks), in both directions: hexafluorobenzene gives no HSQC spots, benzene gives no off-diagonal COSY spots, methane gives none in HMBC.

Declared limits. This is not a simulation of the experiment: no computed intensities, no artefacts, no residual coupling, no mixing-time dependence. In a real HMBC some two-bond correlations are not seen; here they all are. The shifts inherit the predictor's measured uncertainty (0.9 ppm on ^{13}C, 0.06 on ^{1}H).

3.8 3D geometry: built from the graph, not downloaded

`bsi-geom3d.js` builds coordinates by **distance geometry**: a bounds matrix from bonds, valence angles, ring diagonals and van der Waals contacts; initial positions random but **reproducible**, because the generator is seeded from the SMILES; iterative correction of the violations. Aromatic systems are projected onto their plane and linear triples are straightened by rotating the branches as rigid bodies, because those two things a distance alone cannot impose.

The indices are the NMR predictor's, and hydrogens are appended at the end: this is the condition without which the peak ↔ atom ↔ 3D link would light up the wrong atom. The PubChem-fed 3D viewer could not be used for this — an SDF's numbering bears no relation to RDKit's.

Measured against known values (`test_geom3d`, 29 checks): C–C 1.54 Å, C=C 1.34, C≡C 1.20, aromatic 1.39, C–H 1.09; tetrahedral angle 109.5°, aromatic 120.0°, alkyne 180.0°. **In both directions**: benzene planar (0.00 Å), cyclohexane **not** planar (0.34). On real molecules the residual stays under 0.03 Å up to about thirty atoms and reaches 0.31 Å on a 76-atom molecule.

Declared limits. It is not a force field and does not minimise an energy: it minimises geometric violations. It does not choose the most stable conformer and does not handle stereochemistry — an R centre and its enantiomer come out the same.

3.9 From document to working: what is deduced and what is read

`bsi-documento.js` opens the file; `bsi-quesito.js` recognises the data in the text; `bsi-elucida.js` works through them. They are three separate steps **on purpose**, because they fail in different ways and conflating them hides the error: if recognition reads "1715" as an IR band when it was a mass, the working that follows is impeccable and the conclusion is wrong. This is why the panel shows **the table of data read above the working**, each entry with where it came from.

Recognition is by section, not by number. An IR band at 1738 and a mass at 150 are both numbers: telling them apart from the value alone is impossible. The technique's label is found (IR, MS, ^{1}H NMR, ^{13}C NMR) and what follows is read, up to the end of the paragraph or the next label.

What it does not do. There is no language model: there are regular expressions and a spectroscopic rule engine. If the text writes the data in an unforeseen form, the data are **not** read — and the module says so rather than working through half of them. The working goes as far as the rules go: compatible functional groups, fragments, counts. **The final structure is not proposed**, because proposing it would mean guessing; a structure can be written and the program **compares** it with the data, saying which signals fit and which do not.

There is no optical character recognition. A scanned page contains pixels, not letters: the text cannot be read, and the module declares it rather than returning an empty string as if the document were empty. The pages stay visible, and that is where pixel trace extraction takes over.

3.10 The two places where the predictor gives way, measured

Declaring "mean deviation 1.0 ppm" and stopping there hides what one actually needs to know: *where* it is wrong. There are two places, and they are known by measurement.

a) Substituted heteroaromatics — the table is missing, not the method

For benzenes the source gives positional substituent increments; for furan, thiophene, pyrrole and pyridine it **does not**. A substituted heteroaromatic therefore gets the values of the *un*substituted parent, and its β carbons all come out equal when in reality they are not.

Measured on the 23/04/2024 problem (structure `O=CC(=Cc1ccco1)CSC`):

observed	predicted	deviation	which carbon
192.04	192.0	0.04	CHO
150.77	150.0	0.77	furan α-C, substituted
146.05	142.7	**3.35**	furan α-C

observed	predicted	deviation	which carbon
135.55	135.0	0.55	=C
135.05	135.0	0.05	=C
118.55	109.6	**8.95**	furan β-C
112.97	109.6	**3.37**	furan β-C
26.67	34.0	**7.33**	CH_2 between C=C and S
15.72	15.6	0.12	S–CH_3

The two β carbons come out **identical** (109.6) because without positional increments nothing distinguishes them. Transferring the benzene increments to the heteroaromatic ring **was tried**: on the *ipso* carbon it works (151.6 against 150.77 measured), but on the β carbons it makes things **worse** — C3 would go from 3.4 to 5.7 ppm of deviation. It was not done: a generalisation that gets worse exactly where the problem is, is not a generalisation.

b) An sp^3 carbon with TWO α substituents

The increments are derived from **mono**-substituted compounds. Adding two of them on the same carbon counts twice an effect that in fact saturates, and the prediction comes out high. Measured twice, same magnitude and same direction:

molecule	carbon	observed	predicted	deviation
benzyl acetate	OCH_2 (ester O + ring)	66.3	73.8	**7.5**
the problem above	CH_2 (C=C + S)	26.7	34.0	**7.3**

It is the module's declared worst case, and benzyl acetate entered the validation set precisely to keep it measured.

The comparison tool says so. When a proposed structure contains such a carbon, confidence drops from "high" to "medium" and the reason appears: *a signal that does not fit on that carbon may be a limit of the predictor, not evidence against the structure*. Without that warning, a low score on a **correct** structure would read as a refutation — which is exactly what happened to benzyl acetate, which scores 42 out of 100 while being the answer.

3.11 What could be consulted, and what could not

A table of increments for substituted heteroaromatics would close the weak point described above. **From the machine this project is built on, no primary source is reachable**, and that was verified, not assumed:

route	outcome
`curl` to any domain	the network policy allows **only** package registries (npm, PyPI, crates, Go) — the proxy says so itself
NIST WebBook, SDBS (AIST), PubChem, SpectraBase	**403**, blocked

route	outcome
CSIRO, RSC, ACS, Springer, Wikipedia, LibreTexts	**403**, blocked
the Hans Reich tables (`organicchemistrydata.org`, `chem.wisc.edu`)	**403**, blocked
NMRShiftDB2 (`nmrshiftdb.nmr.uni-koeln.de`)	**403**, blocked
PubMed (through its service)	no results: it indexes biomedicine, not pure chemistry
npm and PyPI	no package redistributes a reference NMR database: `nmr-predictor` **downloads** it from GitHub, which is blocked; `nmr-processing` does not contain it

What text search could establish is *where* the data are, and that is a useful result even without having them in hand:

> M. T. W. Hearn, "Carbon-13 chemical shifts in some substituted furans and thiophens", *Australian Journal of Chemistry* **29**(1), 107–113 (1976). DOI 10.1071/CH9760107

The abstract states that "the substituent effects in these heterocycles **resemble** those reported for substituted benzenes". *Resemble* is not *equal*: transferring the benzene increments was tried (§3.10 a) and on the β carbons it makes things worse. The real table is still needed.

Why it was not written from memory. It would have been easy: those numbers are well known. But a value nobody can trace to a source is exactly what this document exists to prevent, and putting one in a table next to ninety-one rows transcribed from a printed page would make those less credible too. **A declared hole is worth more than a badly patched one.**

One validation row with weaker provenance

For honesty this too must be said: benzyl acetate, which entered the validation set because it shows the two-α-substituent weak point, carries values **not verified against a primary source** from this environment. The phenomenon it shows is independently confirmed by the 23/04/2024 exam problem — a datum with real provenance — but the **declared worst case (7.5 ppm) rests on that row**, and the bench says so in plain words.

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4. Physical constants and tabulated data

The lookup of physical constants proceeds by decreasing specificity (exact match → symbol → significant words → substring of at least 4 characters) and **collects every candidate**: if more than one remains, the request is refused with the list of possibilities rather than arbi-

trarily returning the first. A wrong physical value returned with confidence is worse than a request for clarification.

The exam molecules (`MOLECOLE_ESAME`) are tabulated with verified values; the recognition of groups from SMILES uses a `consuma` field, so that each group removes the portion of structure it has explained and no overlapping counts are produced.

4-bis. Astrophysics, physics and mathematics laboratories, study kits

4-bis.1 Where the data come from

Section	Data used	Source declared in the code
🪐 Solar system 3D, 🔭 Planetarium 3D	J2000 Keplerian elements of the planets and their secular rates	E. M. Standish's table (JPL), «Keplerian elements for the approximate positions of the major planets», valid 1800–2050
🪐 Solar system 3D, 🔭 Planetarium 3D	Julian date, sidereal time, Sun, Moon, Galilean moons	J. Meeus, *Astronomical Algorithms* (chapters 7, 12, 25, 44, 47; lunar theory truncated to the main terms, ~0.1°)
🔭 Planetarium 3D	About 170 bright stars: α, δ (J2000), V, B − V; 28 constellations	Rounded values; proper motions are neglected
🌠 Cosmology 3D	Cosmological parameters	Planck 2018: $\Omega_m = 0.315$, $\Omega_\Lambda = 0.685$, $H_0 = 67.4$ km/s/Mpc
🌌 Galaxies 3D	Local Group positions and distances	Compilations by Mateo (1998) and McConnachie (2012), rounded values
⭐ Life of stars 3D, ☢️ Nuclei and reactions	Atomic masses	AME2016/AME2020, rounded
🕳️ Black holes, 🔢 Constants and units	Physical constants	CODATA 2018; those defining the SI since 2019 are marked exact

Declared illustrative, and the sections say so: the exoplanet mass-period population (not a catalogue), the rotation-curve «observations» (the reference model with reproducible noise), the «Milky-Way-like» disc parameters, the cosmic microwave background map (synthesised with spherical harmonics from an illustrative power spectrum, tuned so that the peaks resemble Planck's).

4-bis.2 How it is verified

Each of the twenty new sections has a script in `tools/banchi/verifiche/verifica-<id>.js`, run by `test_v237`, that sets the computations against a known value or an independent second method. Some examples, with the value the script prints:

Section	Comparison	Obtained
🪐 Solar system	Julian date of 1957 Oct 4.81 (Meeus, ex. 7.a)	2436116.31
🪐 Solar system	Mars 2003: closest approach 0.3727 AU on 27 August	0.3730 AU on 27/8
🛰️ Orbital mechanics	Curtis, ex. 4.3: h, i, Ω, e, ω, ν, a from the state	58312, 153.25°, 255.28°, 0.1712, 20.07°, 28.45°, 8788 km
🛰️ Orbital mechanics	elements → state → elements over 2000 random orbits	relative error $2.8 \cdot 10^{-14}$
📐 Celestial mathematics	Laplace limit of the Lagrange series	0.6627434
⭐ Life of stars	Lane–Emden: first zero for n = 3	6.89685 ($\rho_c/\bar{\rho}$ = 54.18)
⚫ Black holes	perihelion precession of Mercury, 42.98″ per century	42.982″
⚫ Black holes	light deflection at the Sun's limb, 1.75″	1.7512″
〰️ Gravitational waves	chirp mass of GW150914 (36 + 29 M☉)	28.096 M☉
🌌 Galaxies	Freeman disc: velocity peak at R/R_d = 2.15	2.150
🌠 Cosmology	age of the universe with Planck parameters (13.8 ± 0.1 Gyr)	13.791 Gyr
🔭 Planetarium	Greenwich sidereal time, 1987 Apr 10 0h UT (Meeus, ex. 12.a)	13.1795463 h
🌍 Exoplanets	Earth's equilibrium temperature (A = 0.3)	254.8 K
🌀 Spin and qubits	HZH = X, HXH = Z	deviation $2.2 \cdot 10^{-16}$
🎲 Statistical mechanics	Schottky peak: kT = 0.4168 Δ, C = 0.4392 k	0.41678; 0.43923
☢️ Nuclei	B/A of ^{56}Fe from the masses	8.7903 MeV
☢️ Nuclei	Q of D + T → ^{4}He + n	17.589 MeV
🧱 Materials	I-section: formula against Huygens–Steiner	7.9990 against 7.9990 ($\cdot 10^{7}$ mm⁴)
🔌 Electronics	diode: Newton against the closed form with Lambert's W	0.650463089 V in both
🌊 PDEs	zero j_{01} of the Bessel function	2.404825557695773
🌀 Complex analysis	e^(iπ) + 1 = 0	$1.2 \cdot 10^{-16}$
🔗 Markov	stationary distribution by powers and by Cramer's rule	π_0 = 0.4615384615384615 in both
🧩 SVD	[[1,1],[0,1]]: $\sigma_1 = \varphi$, $\sigma_2 = 1/\varphi$	1.6180339887498947; 0.6180339887498948
📡 Codes	CRC-32("123456789") = 0xCBF43926; Hamming(7,4) corrects all 112 single errors	0xCBF43926; 112 of 112

Altogether the twenty scripts run **705 checks**. The study-kit scripts check the data schema and minimum quantities and, for mathematics, recompute **264 quiz answers** out of 346 questions; the resources script checks the citation formats on hand-built Crossref records and the tables' numeric proofs.

4-bis.3 The 3D model of Spectra's face

The face of the Spectra Live console is the «Noir» model by Skyfox13 (<https://sketchfab.com/podderkuntal25>), published on Sketchfab under the CC BY 4.0 licence, file `models/spectra-noir.glb`. It has been **modified**: textures recompressed, tangents removed, the face rendered as dark grey metal, head and mouth animated by the code. It carries no scientific data; the attribution appears in the console and in `07-SBOM.md`.

4-ter. The v238 3D tabs

Twenty-nine sections receive a 3D tab built **on the same model** as the section: the scene draws computed quantities (the LEPS surface, the cells with their lattice constants, the integrated field lines, the Lorenz solution, the mean-curvature flow of the soap film…), not hand-made figures. Every tab has a script `tools/banchi/verifiche/verifica-<id>-3d.js` (for Topology and Non-Euclidean geometries the section's own script), run by `test_3d238`; **311 checks** in all, described in full in `15-Test-Documentation.md` §3.1-ter.

Kind of comparison	Examples, with the value the script prints
Measured or published value	the LEPS model's ω(H_2) 4396.6 cm^{-1} against 4401 measured; H_2 at 1.4 bohr −1.116714 Eh (Szabo-Ostlund); acentric factor of water and CO_2 0.3443 and 0.2249
Closed form	Larmor radius 0.61538; h*/R = 1.32549 for the catenoid (u* tanh u* = 1); volume of the sphere 4π/3 with Simpson
Second method	cell lists against the sum over all pairs ($5\cdot10^{-14}$); sphere rotations against SU(2) Möbius maps ($5.5\cdot10^{-16}$); hyperboloid against Poincaré disc ($1.2\cdot10^{-15}$)
Statistical quantity within its error	fraction of collisions above threshold = e^(−E_0/kT) within 3σ; K measured in the equilibrium box within 1.5 %

Declared in the code as models or literature values, not measurements by the app: the critical temperature of the 3D Ising model, T_c ≈ 4.5115 J/k, comes from published simulations (there is no exact solution) and is used as a reference; the detector is «like CMS» (3.8 T solenoid, yoke at −1.8 T, 13 tracker layers up to 1.08 m) with Z → μμ, Z → ee and H → γγ events generated by the code (m_H = 125.2 GeV); the molecular and crystal parameters are those already used and verified by the 2D tabs of the same sections.

Fixed before publication. In molecular collisions the collision normal was taken from spheres the discrete step had already overlapped, and grazing collisions came out more head-on: the fraction above threshold at E_0/kT = 1 was 0.345 against e^{-1} = 0.368. Now the motion is traced back along the relative velocity to the contact: 0.370. On the Riemann

sphere the pole was not recognised for an angle of exactly π, because cos(π/2) in floating point is $6 \cdot 10^{-17}$: now the pole is recognised from the ratio of the Möbius denominator to its numerator.

5. Automation of the verification

All the checks described are runnable and repeatable. At the time of writing the battery includes, among others:

Bench	Subject	Checks
`audit_farmaci`	structure ⟷ weight, duplicates, formula rebuilt from the graph against ChEMBL, deviations recorded with their reason	263 entries
`test_spettri`	group recognition on reference molecules	36
`test_assi` / `test_assi_canvas`	axis conventions (SVG and canvas)	19
`test_costanti`	physical-constant lookup and refusal of ambiguities	45
`audit_dati`	consistency of the tabulated data	29
`verifica_guida`	correspondence between what the documentation promises and what the code does	71
`test_v237`	the twenty astrophysics, physics and mathematics sections against known values or a second method (§4-bis.2), the study kits and the resources	705 in the twenty scripts, plus the kits, the resources and the in-browser checks
`test_3d238`	the 3D tabs of 29 sections against known values, closed forms or a second method (§4-ter), and every tab opened in the browser at 1280 and 390 px	311 in the 29 scripts, 488 in all

The last bench deserves a mention: it verifies that the statements contained in the user documentation correspond to the real behaviour of the code. A promise the documentation does not keep is a defect on a par with a computational error, and is treated as such.

6. Transparency statement

This document describes checks that are **actually implemented and runnable**, with the results actually obtained and the limits of the predictors. At version `bsi-v188` the residual errors on the drug database are **zero**: the 21 deviations that remain are entries without a structure, each with its own recorded reason, not errors passed over in silence.

Where a datum is not verifiable with the instruments available, that is declared rather than presented as verified.

Document updated to version `bsi-v248`.